

Bufan Gao

PhD candidate, Social Psychology, University of Chicago

[Page](#) · [Github](#) · [Scholar](#) · [Mail](#)

1. Research Interests

Multi-agent Systems in Humans and AI; Collective behaviors; Computational Psychology; Social cognition

2. Education

The University of Chicago PhD Student in Social Psychology, advised by Prof. Xuechunzi Bai	2025 — Present
Fudan University Master in Computer Science, advised by Prof. Yuan Zhang and Prof. Jiarun Dai	2022 — 2025
Fudan University B.S. in Department of Computer Science	2018 — 2022

3. Publications

Published:

- **SCTrans: Constructing a Large Public Scenario Dataset for Simulation Testing of Autonomous Driving Systems**
Jiarun Dai*, **Bufan Gao***, Mingyuan Luo, Zongan Huang, Zhongrui Li, Yuan Zhang, Min Yang
In the 2024 International Conference on Software Engineering ([ICSE'24](#))
- **Measuring Bias or Measuring the Task: Understanding the Brittle Nature of LLM Gender Biases**
Bufan Gao, Elisa Kreiss
In the 2025 Conference on Empirical Methods in Natural Language Processing ([EMNLP'25](#))

Under Review / In Preparation:

- **Rational Social Learning Makes Group Hiring More Efficient And Biased**
Bufan Gao, Xuechunzi Bai
Under review in PNAS
- **ExplorIt: Facilitating Simulation-based Fuzzing for Autonomous Driving Systems through Multi-Agent Interaction Modeling**
Bufan Gao, Zongan Huang, Jiarun Dai, Kairui Yang, Yixing Luo, Yingjie Fu, Yuan Zhang, Min Yang, Tao Xie
Under review of the Journal of ACM Transactions on Software Engineering and Methodology ([TOSEM](#))

4. Awards & Grants

- Norman Henry Anderson Travel Award, Winter 2026
- Norman Henry Anderson Research Award, Winter 2026
- PGSO Travel & Research Grant, 2025
- School of Computer Science and Technology Academic Year Scholarship, 2025
- ICSE Student Travel Grant, 2024
- School of Computer Science and Technology Academic Year Scholarship, 2024
- Venustech Scholarship (Top 3% cross department), 2023
- Champion of 2022 DEFCON AutoDriving Contest, 2022
- Fudan University Freshman Scholarship, 2022
- Champion of 2021 DEFCON AutoDriving Contest, 2021
- Fudan University Liu Yong Ling Foundation Scholarship (Top 5% cross university), 2019

5. Talks

- Building Multiplayer Experiments with Humans and LLM Agents, *Tutorial*, [IC2S2 2026](#), Burlington
- Collective Bias Emerges Even from Rational Social Learning, *Paper Talk*, [MPA 2026](#), Chicago
- Rational Social Learning Makes Group Hiring More Efficient and Biased, *Spotlight Data Blitz*, [SPSP 2026](#), Chicago
- Microfoundations of Statistical Discrimination, *Parallel Talk*, [IC2S2 2025](#), Norrköping, Sweden
- Measuring Bias or Measuring the Task: Understanding the Brittle Nature of LLM Gender Biases, *Poster*, [EMNLP 2025](#), Remote
- SCTrans: Constructing a Large Public Scenario Dataset for Simulation Testing of Autonomous Driving Systems, *Main Conference Talk*, [ICSE 2024](#), Lisbon, Portugal

6. Research Experiences

- | | |
|---|--|
| 2024 - Present
Bai Lab | Research Assistant, University of Chicago
Working with Prof. Xuechunzi Bai , I studies collective bias. |
| 2024 - Present
Coalas Lab | Research Assistant, UCLA
While working with Prof. Elisa Kreiss , I studied reliable measurements for bias in large language models (LLMs). |
| 2021 - Present
Secsys Lab | B.S. and Master Student, Fudan University
Under the supervision of Prof. Min Yang, Prof. Yuan Zhang and Prof. Jiarun Dai , with the focus on the <i>Robustness of Autonomous Driving Systems (ADS)</i> . |
| 2021 Summer | Computer Science Summer Research Fellowship Program, University of Chicago
Advised by Prof. Junchen Jiang and Prof. Nick Feamster |

7. Service

- | | |
|---------------------------|------------------|
| ICSE 2026 workshop | Committee Member |
| ICSE 2025 workshop | Committee Member |

8. Teaching

Principles of Reverse Engineering (PoRE), Teaching Assistant

Spring 2023, Spring 2022